

ANDREW SANG-JIN CHOI

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Website <https://quantumope.github.io>

Code <https://github.com/QuantuMope>

Google Scholar <https://tinyurl.com/quantumope>

EDUCATION

Research Interests: **robot learning, sim-to-real, reinforcement learning, simulation**

Doctor of Philosophy, Computer Science

Sept. 2021 – Dec. 2023

Major: Graphics & Vision | Minors: Robotics & Artificial Intelligence

GPA: 4.0/4.0

University of California, Los Angeles

Advisors: M. Khalid Jawed, Jungseock Joo, and Demetri Terzopoulos

Ph.D. Thesis: *Simulation of Deformable Objects for Sim2Real Applications in Robotics*

Master of Science, Computer Science

Sept. 2019 – June 2021

University of California, Los Angeles

GPA: 4.0/4.0

M.S. Thesis: *An Implicit Contact Method for Tying Discrete Elastic Knots*

Bachelor of Science, Mechanical Engineering

Sept. 2014 – June 2018

University of California, Davis

GPA: 3.8/4.0

Senior Design: *Underactuated Shoe Tying Manipulator*

Top 6% in College of Eng.

IN-PROGRESS / PREPRINTS

- CoRL 2026 [A. Choi](#), X. Wang, Z. Su, and W. Xu. “Scaling Sim-to-Real Reinforcement Learning for Robot VLAs with 3D World Generative Models”, in-progress, 2026
- arXiv 2025 [A. Choi](#) and D. Tong. “Rapidly Learning Soft Robot Control via Implicit Time-Stepping”, arXiv, 2025

PUBLICATIONS

- EML 2025 D. Tong*, [A. Choi*](#), J. Wang*, W. Huang*, Z. Chen, J. Li, X. Huang, M. Liu, H. Gao, and K. Jimmy Hsia. “Discrete differential geometry for simulating nonlinear behaviors of flexible systems: A survey”, *Extreme Mechanics Letters*, 2025
- RA-L 2025 H. Zhang, H. Yu, L. Zhao, [A. Choi](#), Q. Bai, Y. Yang, and W. Xu. “Learning Multi-Stage Pick-and-Place with a Legged Mobile Manipulator”, *IEEE Robotics and Automation Letters*, 2025
- NeurIPS 2024 A. Vepa, Z. Yang, [A. Choi](#), J. Joo, F. Scalzo, and Y. Sun. “Integrating Deep Metric Learning with Coreset for Active Learning in 3D Segmentation”, *Conference on Neural Information Processing Systems*, 2024
- IJRR 2024 D. Tong, [A. Choi](#), W. Huang, L. Qin, J. Joo, and M. K. Jawed. “Sim2Real Neural Controllers for Physics-Based Robotic Deployment of Deformable Linear Objects”, *The International Journal of Robotics Research*, 2024
- T-ASE 2024 [A. Choi*](#), D. Tong*, D. Terzopoulos, J. Joo, and M. K. Jawed. “Learning Neural Force Manifolds for Sim2Real Robotic Symmetrical Paper Folding”, *IEEE Transactions on Automation Science and Engineering*, 2024
- RA-L 2024 [A. Choi](#), R. Jing, A. Sabelhaus, and M. K. Jawed. “DisMech: A Discrete Differential Geometry-based Physical Simulator for Soft Robots and Structures”, *IEEE Robotics and Automation Letters*, 2024
- RA-L 2023 [A. Choi](#), D. Tong, B. Park, D. Terzopoulos, J. Joo, and M. K. Jawed. “mBEST: Realtime Deformable Linear Object Detection Through Minimal Bending Energy Skeleton Pixel Traversals”, *IEEE Robotics and Automation Letters*, 2023
- EML 2023 D. Tong*, [A. Choi*](#), J. Joo, and M. K. Jawed. “A Fully Implicit Method for Robust Frictional Contact Handling in Elastic Rods”, *Extreme Mechanics Letters*, 2023
- JAM 2023 D. Tong, [A. Choi](#), J. Joo, A. Borum, and M. K. Jawed. “Snap Buckling in Overhand Knots”, *Journal of Applied Mechanics*, 2023
- ICRA 2022 [A. Choi](#), M. K. Jawed, and J. Joo. “Preemptive Motion Planning for Human-to-Robot Indirect Placement Handovers”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2022

WACV 2022	A. Vepa, A. Choi , N. Nakhaei, W. Lee, et al. “Weakly-Supervised Convolutional Neural Networks for Vessel Segmentation in Cerebral Angiography”, <i>IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)</i> , 2022
JAM 2021	A. Choi , D. Tong, M. K. Jawed, and J. Joo. “Implicit Contact Model for Elastic Rods in Knot Tying”, <i>Journal of Applied Mechanics</i> , 2021

* **Equal Contribution**

PRESENTATIONS

APS 2024	R. Lahoti, A. Choi , and M. K. Jawed. “DiSMech: A Simulator for Soft Robots and Flexible Structures based on Discrete Differential Geometry”, <i>Bulletin of the American Physical Society</i> , 2024
APS 2024	G. Wang, D. Tong, and A. Choi . “Computer Graphics Inspired Fast Simulation of Knots and Tangles”, <i>Bulletin of the American Physical Society</i> , 2024
APS 2023	A. Choi , D. Tong, D. Terzopoulos, J. Joo, and M. K. Jawed. “Deep Learning Force Manifolds from the Physical Simulation of Robotic Paper Folding”, <i>Bulletin of the American Physical Society</i> , 2023
APS 2023	D. Tong, A. Choi , J. Joo, and M. K. Jawed. “An Implicit Simulation Framework to Handle Frictional Contact in Elastic Rods”, <i>Bulletin of the American Physical Society</i> , 2023

INDUSTRY EXPERIENCES

Research Scientist Horizon Robotics, General AI Laboratory Supervisor: Wei Xu	Cupertino, California Feb. 2024 – Present
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- Made significant contributions to the team’s software and hardware infrastructure, including:
 - Developed a robust, general-purpose FK/IK library in C++. Implemented a joint delta-constrained optimization-based IK solver for smooth robot control in RL and teleoperation, achieving over an order-of-magnitude speedup compared to open-source solvers.
 - Built a multi-camera robot hand-eye calibration pipeline.
 - Designed a teleoperation interface using a PS4 controller.
 - Integrated a highly parallelized GPU simulator using ManiSkill 3 with custom improvements (e.g., OMP batched IK solvers). Designed the software API for instantiating training environments that support both RL-in-real settings (reset-free, scripted resets, reward models) and large-scale parallel RL training.
 - Wired and installed a soft gripper on a Unitree Z1 manipulator via a custom 3D-printed adapter. Implemented C++ drivers for controlling both the gripper and the Z1.
 - Built a rapid, parallelized data transfer pipeline for large-scale upload/download between the local network and GPU cluster.
 - Contributed regular commits to the team’s RL framework (alf).
- Developed sim-to-real teacher–student policies for long-horizon legged mobile manipulation. Contributed substantially to RL tuning and sim-to-real transfer, resulting in a publication in RA-L 2025.
- Trained robot manipulator policies using hybrid RL with VLAs as priors. Collected an offline dataset of successful trajectories using OpenVLA and leveraged it as an offline buffer with RLPD, enabling from-scratch learning of visual pick-and-place in a few hours. Subsequently pivoted to direct VLA fine-tuning.
- Processed and curated large open-source robot datasets (e.g., OXE) for training VLM-based (PaliGemma2) reward models, which were used to train RL policies in both simulation and real-world settings.
- Implemented state-of-the-art RL fine-tuning algorithms for flow-matching models, including Diffusion Steering RL (DSRL), Flow Policy Optimization (FPO), PPOFlow, and SACFlow.
- Fine-tuned multiple VLAs, including OpenVLA (via PPO) and π_0 (via DSRL, FPO, PPOFlow, and SACFlow). Deployed additional models, including Octo and SpatialVLA.
- Led a first-author project combining 3D generative world models with highly parallelized GPU simulation for scaling sim-to-real RL fine-tuning of VLAs. Manuscript in preparation with a target submission to CoRL 2026.
- Received performance ratings of **Consistently Exceeding Expectations** (top 30%) in year one and **Outstanding** (top 10%, highest rating with company-wide announcement) in year two.

Graduate Robotic Software Intern Vecna Robotics, Research & Advanced Development Division Supervisors: Siddharth Chhaptar & Magnus Snorrason	Waltham, Massachusetts Summer of 2021
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- Created an end-to-end robotic solution for autonomous refueling of self-driving warehouse vehicles using a 4DOF serial robot manipulator. Achieved a 99.6% success rate for a client specification of 98%.
- Created hydrogen fuel cell nozzle pose estimation and plane segmentation module using Aruco and RANSAC.
- Carried out motion trajectories using velocity control and inverse Jacobian method for the inverse kinematics.
- Designed the control flow pipeline, control & recovery states, and algorithms for the refueling action.
- Created autonomous testing pipeline and GUI for the autonomous refueling framework with the robot arm and mobile vehicle communicating to each other through server connection.
- Researched and implemented an open-source eye-in-hand hand-eye calibration module using dual quaternions for 4DOF manipulators.
- Created URDF models through Xacro for omnidirectional mobile forklift.
- Derived omnidirectional kinematics and created ROS plugin for Gazebo simulations. Solved inverse kinematics using a constraint-based optimization approach.

Control Systems Engineer

Brock Solutions

Los Angeles, California

Nov. 2018 – July 2019

- Programmed and debugged the programmable logic controllers (PLCs) that control the various actuators and sensors of the automated LAX baggage handling systems.
- Maintained human machine interfaces that display live system status in a high-stakes real time environment.
- Interfaced with real-time database applications to manage critical production information for airline clients.

R&D Engineering Intern

World Surf League (Kelly Slater Wave Company)

Advisor: Adam Fincham, USC

Culver City, California

Summer of 2017

- Collected and stored data from test runs that led to the world's first artificial 6-foot barreling surf wave.
- Installed, programmed, and maintained pressure sensors, strain sensors, cameras, drones, and velocimeters.
- Performed force calculations for safety structures and managed the ordering of necessary materials/tools.

ACADEMIC & TEACHING EXPERIENCES

Graduate Student Researcher

UCLA, Structures-Computer Interaction Lab

Advisors: Mohammad Khalid Jawed & Jungseock Joo & Demetri Terzopoulos, UCLA

Los Angeles, California

Jan. 2020 – Dec. 2023

- Perform research in the general intersection of robotics, simulations, and learning with a focus on sim-to-real problems.
- Key research areas include #1) developing physically accurate simulations for deformable materials / soft robots and #2) developing sim-to-real solutions for robotic manipulation of deformable materials.

Teaching Assistant for CS32: Introduction to Computer Science II (In-Person)

UCLA, Computer Science Department

Instructor: David Smallberg, UCLA

Los Angeles, California

Jan. 2023 – March 2023

- Taught students how to employ object-oriented programming to build large programs as well as a variety of fundamental data structures and algorithms in C++.
- Held office hours and led discussion sessions consisting of 43 students.
- Received a TA evaluation score of 8.0/9.0 with 19/43 (44.2%) survey participation.

Teaching Assistant for CS31: Introduction to Computer Science I (Online)

UCLA, Computer Science Department

Instructor: Howard Stahl, UCLA

Los Angeles, California

Summer of 2022

- Taught students fundamental concepts and principles of CS as well as general programming principles in C++.
- Held office hours and led discussion sessions consisting of 85 students.
- Received a TA evaluation score of 8.6/9.0 with 9/85 (10.6%) survey participation.

Robotics Senior Design Presentation

Meijo University, Robotics Department

Advisor: Jason K. Moore, UC Davis (now at TU Delft)

Nagoya, Japan

Summer of 2018

- Fully funded trip to present the design theory and findings of underactuated shot tying robot manipulator to robotics students and professors of Meijo University.
- Acted as a mentor to students completing their own robotics senior design projects. Provided technical guidance and feedback and partook in culture exchange programs.

Undergraduate Researcher
UC Davis, Green Technology Lab
Advisor: Masoud Rahman, UC Davis

Davis, California
April 2017 – June 2018

- Performed economic analysis on different energy management techniques for various battery systems.
- Modeled the energy parameters of residential and commercial communities using behavioral survey data, load data, PV generation data, etc. to calculate optimal battery sizing.

HONORS & ACHIEVEMENTS

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|---|-----------------------|
| 1. Senior Design Robotics Television Broadcast | CBS/NBC, 2018 |
| 2. Meijo University Travel Grant | UC Davis, 2018 |
| 3. Magna Cum Laude <i>Top 8% in College of Eng.</i> | UC Davis, 2018 |
| 4. UC Davis / Meijo University Annual Robotics Competition Winner | UC Davis, 2018 |
| 5. Engineering Dean's List <i>Top 16% in College of Eng. 10 out of 12 quarters</i> | UC Davis, 2014 - 2018 |
| 6. H. and G. Smith Undergraduate Scholarship <i>Based on scholastic potential</i> | UC Davis, 2015 |

SKILLS

Programming Languages	C/C++, Python, Java, MATLAB
Software Tools and Environments	ROS, Linux, Bash, Git/Github, L ^A T _E X, Unity
Libraries	NumPy, OpenCV, MoveIt, SciPy, SymPy, PyTorch, TF2/Keras
Prototyping	Arduino, Beaglebone, Raspberry Pi, 3D Printing, SOLIDWORKS
Spoken Languages	English (Native), Korean (Heritage), Spanish (Limited Working Proficiency)

COURSEWORK

Math & Probability: Linear Algebra, Differential Equations, Probabilistic Systems Analysis

Robotics & Engineering: Electrical Circuits & Systems, Dynamics, Mechanical Design, Automatic Control of Engineering Systems, Simulation & Design of Mechatronic Systems, Measurement Systems, Vehicle Stability, Kinematics of Robotic Systems, Computational Robotics, Deformable Simulations for Soft Robotics

Computer Science & Machine Learning: Data Structures, Software Construction, Computer Organization, Operating Systems Principles, Programming Languages, Computer Network Fundamentals, Algorithms & Complexity, Machine Learning Algorithms, Machine Learning in Genetics, Pattern Recognition, Deformable Models in Computer Vision, Reinforcement Learning, Artificial Life in Computer Graphics & Vision, Human-centered AI